Wireshark Packet Analysis & OSI Model Assignment

Comprehensive Long-Form Technical Report & Detailed Solutions (Tasks 1 to 4)

Task 1: Wireshark Packet Capture & Protocol Identification (Visiting Instagram)

Wireshark is a powerful open-source packet analyzer that captures live network traffic passing through a network interface card (Wi-Fi or Ethernet adapter). When launching a capture while browsing a secure HTTPS website like `instagram.com`, Wireshark logs hundreds of frames traversing different protocol layers. Below is an overview of the documented capture session and observed protocols:

Step-by-Step Procedure & Observed Protocols:

- **1. Initialization:** Wireshark was installed, and live capture was initiated on the primary wireless adapter (`Wi-Fi / en0`).
- **2. Web Traffic Generation:** Opened a web browser and navigated to `https://www.instagram.com`, allowing the page elements, images, and API JSON scripts to load fully.
- **3. Capture Termination:** Stopped the Wireshark capture and applied filters (e.g., `tls` or `http`) to examine traffic.

Key Protocols Observed in Wi-Fi Capture List:

Protocol Name	OSI Layer	Functional Role during Instagram Browsing Session
DNS (Domain Name System)	Application (Layer 7)	Resolves human-readable domain `instagram.com` into Meta's IP addresses.
TLS (Transport Layer Security)	Presentation / Transport	Establishes a secure, encrypted HTTPS tunnel for all user data and photo transfers.
TCP (Transmission Control Protocol)	Transport (Layer 4)	Manages reliable, ordered packet delivery with handshakes (SYN, ACK) and flow control.
QUIC / HTTP/3	Transport / Application	Modern UDP-based transport protocol used by Meta for ultra-fast media streaming.
ARP (Address Resolution Protocol)	Data Link (Layer 2)	Maps local gateway IP addresses to physical MAC addresses on the local Wi-Fi LAN.

Task 2: Packet Mapping to OSI Layers

To understand how network communication works, we select three distinct packets from the Wi-Fi capture log and map them to their corresponding OSI layers based on their protocol headers and Info descriptions:

Packet #	Observed Protocol	OSI Layer Mapping	Wi-Fi Capture Info & Analysis
Packet #12	**DNS**	**Application Layer (Layer 7)**	*Info:* Standard query A `instagram.com`. The browser requests the IP address for Instagram from the local DNS resolver.

Packet #	Observed Protocol	OSI Layer Mapping	Wi-Fi Capture Info & Analysis
Packet #15	**TCP**	**Transport Layer (Layer 4)**	*Info:* [SYN] packet sent to port 443 of Instagram server. Establishes a reliable connection socket before TLS handshake.
Packet #45	**TLSv1.3**	**Presentation / Application Layer**	*Info:* Encrypted Handshake / Application Data. Negotiates cryptographic keys and securely transmits encrypted Instagram feed media.

Task 3: OSI Model Reference Diagram & Real-World Daily App Examples

The Open Systems Interconnection (OSI) model conceptualizes telecommunication and computer networking into seven distinct abstraction layers. Below is a structured reference table mapping each OSI layer to real-world applications and services used daily:

Layer #	OSI Layer Name	Core Function & Responsibility	Real-World Daily App Example
Layer 7	**Application Layer**	Provides network services directly to end-user software applications.	**WhatsApp / Gmail:** Sending text messages or composing emails.
Layer 6	**Presentation Layer**	Handles data formatting, encryption, compression, and character encoding.	**Instagram / Netflix:** Decoding JPEG/PNG images, WebP photos, and H.264 video codecs.
Layer 5	**Session Layer**	Establishes, maintains, and terminates communication sessions between apps.	**Zoom / Google Meet:** Maintaining active video call authentication and synchronization.
Layer 4	**Transport Layer**	Ensures reliable, error-checked delivery of data segments (TCP/UDP).	**Online Multiplayer Games / YouTube:** TCP for reliable chat, UDP for live video buffering.
Layer 3	**Network Layer**	Handles logical IP addressing and routing packets across multiple routers (IP).	**Google Maps Navigation:** Routing GPS data packets across the global internet.
Layer 2	**Data Link Layer**	Manages physical node-to-node data transfer and MAC address framing.	**Home Wi-Fi Router / Ethernet Switch:** Framing data between your laptop and Wi-Fi AP.
Layer 1	**Physical Layer**	Transmits raw electrical voltages, light pulses, or RF radio waves over physical media.	**5G Mobile Data / Home Wi-Fi Antennas:** Radiating $2.4 ext{ GHz} / 5 ext{ GHz}$ radio waves through the air.

Task 4: Packet Journey: Sending a Message on WhatsApp through 4 Layers of the OSI Model

When you type and send a chat message (e.g., *"Hello!"*) to a friend on **WhatsApp**, your smartphone encapsulates and processes the data downward through the OSI layers before transmitting it into the airwaves. Below is the step-by-step traversal across 4 key OSI layers:

Step 1: Application Layer (Layer 7) — Message Creation & UI Interaction

You open WhatsApp, type `"Hello!"` in the chat box, and tap the send icon. The WhatsApp application software generates the message content and formats it into application-layer data packets using protocols like MQTT or HTTPS (WebSockets) designed for real-time messaging.

Step 2: Transport Layer (Layer 4) — Segmentation & Reliable Delivery (TCP)

The operating system's networking stack takes the WhatsApp message data, chops it into manageable chunks (segments), and adds a **TCP header**. TCP assigns sequence numbers and sets up port numbers (e.g., HTTPS port 443) to guarantee that your message arrives reliably, in the correct order, with error-checking and retransmission if any packet drops.

Step 3: Network Layer (Layer 3) — IP Addressing & Routing

The Transport segment is handed down to the Network Layer, which wraps it into an **IP Packet**. Here, your phone's local private IP address (e.g., `192.168.1.15`) is added as the source, and WhatsApp's cloud server public IP address is added as the destination. Routers across the internet inspect this IP header to forward the packet across the optimal path.

Step 4: Physical Layer (Layer 1) — RF Radio Wave Transmission

Finally, the fully encapsulated digital packet is handed down to the Physical Layer. The phone's internal Wi-Fi antenna converts the digital bits (1s and 0s) into **Radio Frequency (RF) electromagnetic waves** ($ ext{IEEE } 802.11$ standards on $5 ext{ GHz}$ band), radiating them through the air into your home Wi-Fi router, which forwards it onward to WhatsApp's cloud server!